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Direct Computation of Nonlinear Acoustic Pulses Using High-Order Finite Difference Schemes†

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Abstract

The feasibility of using high-order finite difference schemes to compute linear and nonlinear acoustic pulse solutions is investigated. High-order finite difference schemes are generally less dispersive, less dissipative and more isotropic than low-order schemes. Furthermore, these schemes tend to provide wave speeds which are excellent approximations of those of the original governing partial differential equations. For these reasons, they are well suited for solving linear acoustic wave propagation problems. However, it is well-known that for nonlinear problems, high-order finite difference schemes, invariably, generate spurious spatial oscillations. These oscillations emanate from regions of steep gradients and shocks. The presence of these spurious oscillations inevitably renders the computed solution totally unacceptable. The origin and the dominant wavelengths of these spurious oscillations are studied. It is found that the wavelengths are concentrated in a narrow band in the short wave range of the finite difference scheme. The dominant wavelengths are determined primarily by the size of the finite difference stencil. They are only mildly affected by the initial shape of the acoustic pulse and whether the equations are casted in a conservation or nonconservation form before discretization. Since the wavenumbers of the spurious oscillations lie in the short wave range of the finite difference scheme, it is proposed that selective artificial damping terms be added to the difference equations to damp out these short wave components while at the same time leaving the long waves unaffected. In most problems, strong selective damping is needed only in regions of large velocity gradients. To make the artificial damping terms most effective it is further suggested that a variable damping coefficient be used. Ideally, the magnitude of the artificial damping terms should be chosen so that the damping rate just matches the rate of production of the short waves by the nonlinear steepening terms. A measure of the ratio of the magnitudes of these two effects is given by the stencil Reynolds number. An exact match of these effects implies a universal value for the stencil Reynolds number. This value depends only on the finite difference scheme and the damping curve of the artificial damping terms. It is, however, independent of the acoustic pulse problems to be solved. Results of extensive numerical experiments on nonlinear acoustic pulse propagation using artificial selective damping will be reported. It will be demonstrated that for a seven-point stencil and a stencil Reynolds number of 0.1 the computed waveforms are almost free of spurious oscillations. These waveforms compare well with exact solutions.

1. Introduction

Aircraft noise or aeroacoustics problems such as jet noise or propeller noise are governed by the linearized Euler equations outside the noise source region. These problems usually consist of acoustic waves of a large spectrum of wavelengths. To be able to perform numerical simulation of the entire wave spectrum, including even the short waves, computation schemes which are capable of resolving waves with as few as five to six mesh points per wavelength are needed. Further, acoustic waves are nondispersive, isotropic and almost nondissipative. Numerical schemes which are least dispersive, least dissipative and least anisotropic are desirable. Under special conditions, however, additional strong acoustic pulses are radiated from the noise source. This occurs, for example, in the case of screeching jets[1,2] and in the near field of high-speed propfans[3,4] The intensity of these pulses are often high enough for the waveform to be distorted by nonlinear efforts. Even the formation of weak shocks has been observed. Thus, to be able to simulate numerically this sub-class of problems accurately, numerical schemes with shock-capturing capability are favored.

It is generally recognized that high-order finite dif-

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ference schemes are better suited for the computation of linear waves. These schemes, when properly formulated, are less dispersive, less dissipative and more isotropic than low-order schemes. They can reproduce the wave speed accurately using as few as five or six mesh points per wavelength. An example of these schemes is the Dispersion-Relation-Preserving (DRP) scheme of Tam and Webb[5]. To illustrate this point, let us consider the numerical solution of the convective wave equation

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0 \tag{1.1}$$

with initial condition

$$u(x,0) = 0.5 \exp\left[-\ln 2\left(\frac{x}{3}\right)^2\right] \tag{1.2}$$

The length, velocity and time scales are Δx (the mesh size), a_0 (the ambient sound speed) and $\Delta x/a_0$ respectively. Figures 1a to 1d show the computed solution at $t = 400$ using different finite difference schemes. Figure 1a shows the numerical solution based on using the standard second order central difference scheme to approximate the spatial derivative. Obviously, the computed solution differs considerably from the exact solution. The main reason is numerical dispersion. Figures 1b and 1c show the results using the standard fourth and sixth order central difference approximation. Clearly, the computed result improves as the order of the scheme increases. Figure 1d is the computed result using the 7-point stencil DRP scheme. The numerical values of the coefficients of the scheme are provided in section 2 (the DRP scheme optimized over the wavenumber range of $-1.1 \leq \alpha\Delta x \leq 1.1$ improves the numerical group velocity over the standard sixth order scheme). The pulse as prescribed has a very narrow half-width and hence contains a significant fraction of high wavenumber components. The group velocities[6,7,8] of the low-order scheme at these high wavenumbers vary with wavenumber and are not identical to that of the partial differential equation. This results in numerical dispersion.

High-order schemes, on the other hand, are known to be less suitable for nonlinear wave computation especially when shocks are present. The computed results shown in figures 2 and 3 are symptomatic of the problems encountered when using a high-order finite difference scheme to solve nonlinear acoustic wave problems. The computed results are for the nonlinear simple wave equation

$$\frac{\partial u}{\partial t} + \left(1 + \frac{\gamma+1}{2}u\right)\frac{\partial u}{\partial x} = 0 \tag{1.3}$$

where $\gamma = 1.4$. The initial condition used is,

$$u(x,0) = 0.5 \exp\left[-\ln 2\left(\frac{x}{5}\right)^2\right] \tag{1.4}$$

In figure 2a the computed solution using the 7-point DRP scheme and the exact solution at $t = 6$ are shown. The waveform has been severely distorted by nonlinearity. This is right before the formation of a weak shock at the front part of the pulse. The numerical solution matches well with the exact solution. Figure 2b shows the computed waveform and the exact solution (with fitted shock) at $t = 16$. Clearly the numerical solution is dominated by spurious oscillations before and after the shock. In the computed solution, the shock is represented by a steep gradient. As an approximation to the exact solution the numerical solution is not acceptable. Figure 3a and 3b show the computed and the exact solutions of the same problem. The difference is that the governing equation (1.3) has now been rewritten in the conservation form,

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} + \frac{\gamma+1}{4}\frac{\partial u^2}{\partial x} = 0 \tag{1.5}$$

before discretization. A careful comparison between figures 2 and 3 reveals only minor differences in the computed results. This is true for all the other examples we have tested. There appears to be no significant difference between whether a conservation form of the equation is used or not. The same spurious spatial oscillations persist in the numerical solution.

Currently, there are many well established shock-capturing computational fluid dynamics (CFD) schemes in the literature, for example, the second order TVD scheme of Roe[9,10] and Sweby[11], the upwind TVD scheme of Harten[12] and Yee *et.al*[13], the ENO scheme of Harten and Osher[14], the MUSCL scheme of Van Leer[15], Goodman and LeVeque[16], Colella[17] and Davis[18], the PPM scheme of Colella and Woodward[19]; just to name a few. Many of these schemes are reviewed in the recent papers of Yang and Przekwas[20] and Arminjou[21]. Most shock-capturing schemes are low-order schemes with substantial built-in damping. Roe[22] in a recent paper examined the appropriateness of using some of these schemes for linear acoustic waves computation. He gave examples indicating that the strong built-in damping could effectively eliminate the acoustic waves after propagating over the distance of only a few wavelengths.

High-order finite difference schemes which work well for linear acoustics problems are not necessarily suitable for nonlinear wave problems especially

when shocks are formed. On the other hand, shock-capturing schemes which generate crispy shocks are generally low-order schemes. They are not designed or suitable for long distance wave propagation computations. The purpose of this paper is two-fold. First, we like to seek a better understanding of why spurious pre-shock and post-shock spatial oscillations are formed in high-order finite difference solutions. The second objective is to make use of this new understanding to try to devise a way to eliminate the pre- and post-shock oscillations without materially affecting the acoustic pulse. In this way, a high-order finite difference scheme may be used effectively to simulate numerically the propagation of linear as well as nonlinear acoustic waves. Throughout this paper, the DRP scheme will be used as a model of high-order finite difference schemes. The conclusions found should, however, be applicable to similar high-order schemes. It is important to point out, it is not our expectation that the improved high-order schemes would produce crispy shocks as well as the established and proven CFD shock-capturing schemes. Rather, our aim is to modify these schemes so as to be capable of generating reasonable shock profiles with as little pre-shock and post-shock oscillations as possible.

2. Computation of Nonlinear Acoustic Waves Using High-Order Finite Difference Scheme

2.1 Characteristics of high-order schemes

In a recent work, Tam and Webb[5] developed a high-order Dispersion-Relation-Preserving (DRP) finite difference scheme for computational acoustics. The scheme has the unusual property that for long waves (waves with wavelengths longer than five or six mesh spacings) the linear dispersion relations of the finite difference equations are, for all intents and purposes, identical to that of the original partial differential equations. This guarantees that the characteristics of the acoustic waves, the entropy waves and the vorticity waves supported by the finite difference scheme are excellent approximations of those of the linearized Euler equations. In order to preserve the dispersion relations, the spatial and time derivatives are not approximated by finite differences formed in the usual way, namely, by using truncated Taylor series expansion. Rather the approximation is made in the wavenumber space. For instance, consider the approximation of the first derivative $\partial f/\partial x$ at the ℓth node of a uniform grid of spacing Δx using a $2N+1$ point stencil; i.e.,

$$\left(\frac{\partial f}{\partial x}\right)_\ell \simeq \frac{1}{\Delta x}\sum_{j=-N}^{N} a_j f_{\ell+j} \tag{2.1}$$

Equation (2.1) is a special case of the following finite difference equation in which x is a continuous variable:

$$\frac{\partial f}{\partial x}(x) \simeq \frac{1}{\Delta x}\sum_{j=-N}^{N} a_j f(x + j\,\Delta x) \tag{2.2}$$

(2.1) can be recovered from (2.2) by setting $x = \ell\,\Delta x$. The Fourier transform and its inverse of a function are related by:

$$\widetilde{f}(\alpha) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)e^{-i\,\alpha\,x}dx \tag{2.3a}$$

$$f(x) = \int_{-\infty}^{\infty} \widetilde{f}(\alpha)e^{i\,\alpha\,x}d\,\alpha \tag{2.3b}$$

The Fourier transform of the left and right side of (2.2) are:

$$i\,\alpha\,\widetilde{f} \simeq \left(\frac{1}{\Delta x}\sum_{j=-N}^{M} a_j e^{ij\,\alpha\,\Delta\,x}\right)\widetilde{f} \tag{2.4}$$

By comparing the two sides of (2.4) it is clear that the quantity

$$\overline{\alpha} = \frac{-i}{\Delta x}\sum_{j=-N}^{M} a_j e^{ij\,\alpha\,\Delta\,x} \tag{2.5}$$

is effectively the wave number of the Fourier transform of the finite difference scheme (2.2) or (2.1). $\overline{\alpha}\,\Delta\,x$ is a periodic function of $\alpha\,\Delta\,x$ with period 2π. To assure that the Fourier transform of the finite difference scheme is a good approximation of that of the partial derivative over the range of wave numbers of interest it is required that a_j be chosen to minimize the integrated error E defined below:

$$\begin{aligned} E &= \int_{-\eta}^{\eta} |\,\alpha\,\Delta\,x - \overline{\alpha}\,\Delta\,x|^2 d(\alpha\,\Delta\,x) \\ &= \int_{-\eta}^{\eta} \left| i\,\kappa - \sum_{j=-N}^{M} a_j e^{ij\,\kappa}\right|^2 d\,\kappa \end{aligned} \tag{2.6}$$

If the waves of interest have wavelengths longer than $4\Delta x$ or $|\alpha\Delta x| < \pi/2$, then η of (2.6) is set equal to

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$\pi/2$. However, η may be adjusted to further improve the propagation characteristics of the numerical wave spectrum. The conditions that E is a minimum are:

$$\frac{\partial E}{\partial a_j} = 0, \qquad j = -N \text{ to } N \tag{2.7}$$

(2.7) provides a system of linear algebraic equations by which the coefficients a_j can be easily determined.

It is possible to combine the traditional truncated Taylor series finite difference approximation and the wavenumber space approximation described above. For instance, if $N = 3$ one can impose the condition that (2.2) be accurate to order $(\Delta x)^4$. This can be accomplished by expanding the right side of (2.2) by Taylor series in Δx and choose a_2 and a_3 so that terms to order $(\Delta x)^4$ are balanced. This leaves one of the coefficients, say a_1, as a free parameter. This parameter can then be chosen to minimize the error integral E of (2.6). When this constrained minimization is carried out, it is straightforward to find that the coefficients (with $\eta = 1.1$) are:

$$\begin{aligned} a_0 = 0, \quad a_1 &= -a_{-1} = 0.770882380518 \\ a_2 &= -a_{-2} = -0.166705904415 \\ a_3 &= -a_{-3} = 0.020843142770 \end{aligned}$$

(the choice of $\eta = 1.1$ instead of $\pi/2$ as in ref. [5] appears to give the best overall numerical wave characteristics for a 7-point stencil as judged by the $d\overline{\alpha}/d\alpha$ curve). Figure 4 shows the relation $\overline{\alpha}\,\Delta x$ versus $\alpha\,\Delta x$ over the interval 0 to π using the above coefficients. For $\alpha\,\Delta x$ up to 1.2 the curve is nearly the same as the straight line $\overline{\alpha} = \alpha$. Thus the finite difference scheme can provide an adequate approximation to the partial derivative for waves with wavelengths longer than 5 mesh spacings. For $\alpha\,\Delta x$ greater than 1.2 the $\overline{\alpha}(\alpha)$ curve deviates increasingly from the straight line relationship. Because of this the wave propagation characteristics of the short wave components of the finite difference equations would be very different from those of the partial differential equations. We will refer to the wave spectrum for which $\alpha\Delta x > 1.2$ as the short waves and the rest of the spectrum the long waves. The value of $\alpha\Delta x$ at which the $\overline{\alpha}\,\Delta x$ versus $\alpha\Delta x$ curve starts to deviate from the linear relationship will be denoted as $\alpha_c\Delta x$.

Figure 5 shows the $\overline{\alpha}\,\Delta x$ versus the $\alpha\Delta x$ relations for the standard six order, fourth order and second order central difference schemes. It is easy to see that for the sixth order scheme the curve starts to deviate from the linear relation $\overline{\alpha} = \alpha$ at $\alpha\Delta x = 1.0$. In other words, the optimized DRP scheme improves, at no extra cost of computing, the range of resolution by about 10–20%. A better assessment of the improvement of the 7-point DRP scheme over the standard sixth order scheme is to compare the $d\overline{\alpha}/d\alpha$ curves shown in figure 6. The value $d\overline{\alpha}/d\alpha$ should be equal to unity for perfect approximation. It has the meaning of group velocity (for the numerical solution) if the time calculation is exact. Deviation from $d\overline{\alpha}/d\alpha = 1$ produces numerical dispersion.

It is possible to reduce the range of the short waves by increasing the size of the stencil or N. Figure 7 shows the $\overline{\alpha}\,\Delta x$ versus $\alpha\Delta x$ relations for the $N = 3$, 5 and 7 DRP schemes. Obviously, schemes with larger N or stencil can better resolve waves with shorter wavelengths ($\alpha_c\Delta x$ increases with N). This, of course, is at the expense of CPU time. Also, the improvement in resolution in terms of the number of mesh points per wavelength becomes less significant and less impressive when N is larger than 3. In this work, a 7-point stencil ($N = 3$) will be used unless otherwise stated.

In the work of Tam and Webb[5], time is advanced by a four-level finite difference scheme. The coefficients are again optimized so that the Laplace transform variable of the finite difference scheme, $\overline{\omega}$, is a good approximation of the Laplace transform variable ω. With a high-order time marching scheme, there will be spurious transient solutions. However, by a proper choice of Δt, it can be shown that these spurious transient solutions are heavily damped. The computed solution is stable and will faithfully reproduce a good approximation of the solution of the original PDE for all time.

2.2 Origin and characteristics of the spurious oscillations

Now let us apply the DRP scheme to nonlinear acoustic wave problems. The Euler equations in one dimension are:

$$\begin{aligned} \frac{\partial \rho}{\partial t} + u\frac{\partial \rho}{\partial x} + \rho\frac{\partial u}{\partial x} &= 0 \\ \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + \frac{1}{\rho}\frac{\partial p}{\partial x} &= 0 \\ \frac{\partial p}{\partial t} + u\frac{\partial p}{\partial x} + \gamma p\frac{\partial u}{\partial x} &= 0 \end{aligned} \tag{2.8}$$

For acoustic waves propagating into a region at rest, usually referred to as nonlinear simple waves, the above system of equations reduces to a single equation in u by the method of characteristics (see Whitham[23]). The nonlinear simple wave equation is

$$\frac{\partial u}{\partial t} + \left(a_0 + \frac{\gamma+1}{2}u\right)\frac{\partial u}{\partial x} = 0 \tag{2.9}$$

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where a_0 is the speed of sound in the undisturbed region ahead of the wave and γ is the ratio of the specific heats of the gas. For this class of problems, the general initial value problem consisting of equation (2.9) and the initial condition,

$$t = 0, \qquad u = f(x) \tag{2.10}$$

may be solved easily by the kinematic wave method (see Whitham[23]). With nonlinear steepening effects, the solution often contains shocks. The location of the shocks can be readily determined by the equal area rule of Whitham. For simplicity, we will first focus our attention on determining the source of the difficulties in computing the solution of the nonlinear simple wave problem (equation (2.9) and initial condition (2.10)) using a high-order finite difference scheme.

On following the DRP scheme, the discretized forms of (2.9) and (2.10) in dimensionless variables, are (length scale $= \Delta x$, velocity scale $= a_0$, time scale $= \Delta x / a_0$):

$$K_\ell^{(n)} = -\left(1 + \frac{\gamma+1}{2} u_\ell^{(n)}\right) \sum_{j=-3}^{3} a_j u_{\ell+j}^{(n)} \tag{2.11}$$

$$u_\ell^{(n+1)} = u_\ell^{(n)} + \Delta t \sum_{j=0}^{3} b_j K_\ell^{(n-j)} \tag{2.12}$$

$$u_\ell^{(0)} = f(\ell \Delta x); \quad u_\ell^{(n)} = 0 \quad (n < 0) \tag{2.13}$$

where ℓ is the spatial index, n is the time index and Δt is the (dimensionless) time step.

Consider an initial pulse with a Gaussian profile,

$$f(x) = h \exp\left[-\ln 2 \left(\frac{x}{b}\right)^2\right]. \tag{2.14}$$

The parameters h and b are the maximum amplitude and half-width of the pulse respectively. Figure 2a shows the computed pulse shape with $h = 0.5$, $b = 5.0$ at $t = 6$. The waveform, unlike the initial shape, is unsymmetric and highly distorted by the nonlinear steepening effect. The computed result agrees well with the exact solution. $t = 6$ is nearly equal to the wave breaking time; just prior to the formation of a shock. Figure 2b shows the waveform at $t = 16$. At this time a shock has been formed in the exact solution. Clearly, the high-order finite difference scheme cannot reproduce a sharp discontinuity. It is replaced by a steep gradient. The most objectionable features of the finite difference solution are the spurious spatial oscillations ahead and behind the shock. Their presence renders the computed solution totally unacceptable.

What causes the spurious spatial oscillations? A clear answer to this question does not seem to be available in the literature. It turns out a good deal of understanding of the origin of the spurious oscillations can be found by studying and comparing the time evolution of the exact and finite difference solutions in wavenumber space. The Fourier transform of equation (2.9), in dimensionless form, is

$$\frac{d\,\widetilde{u}(\alpha,t)}{dt} + i\left[\alpha\,\widetilde{u}(\alpha,t) + \frac{\gamma+1}{2} \int_{-\infty}^{\infty} k\,\widetilde{u}(k,t)\,\widetilde{u}(\alpha-k,t)dk\right] = 0 \tag{2.15}$$

The initial condition, given by the Fourier transform of (2.10), is

$$\widetilde{u}(\alpha,0) = \widetilde{f}(\alpha) \tag{2.16}$$

For the Gaussian pulse of (2.14) the function $\widetilde{f}(\alpha)$ is

$$\widetilde{f}(\alpha) = \left(\frac{\pi}{\ln 2}\right)^{1/2} hb \exp\left[-\frac{(\alpha\, b)^2}{4 \ln 2}\right].$$

By applying the differencing scheme of (2.1) to the dimensionless form of (2.9), it is straightforward to find that the equation for u at the ℓth mesh point is

$$\frac{du_\ell}{dt} + \left(1 + \frac{\gamma+1}{2} u_\ell\right) \sum_{j=-3}^{3} a_j u_{\ell+j} = 0 \tag{2.17}$$

This is a special case of the following finite difference-differential equation in which x is a continuous variable.

$$\frac{du(x,t)}{dt} + \left(1 + \frac{\gamma+1}{2} u(x,t)\right) \cdot \sum_{j=-3}^{3} a_j u(x + j\Delta x, t) \tag{2.18}$$

(2.17) can be recovered from (2.18) by letting $x = \ell \Delta x$. The Fourier transform of (2.18) is

$$\frac{d\,\widehat{\widetilde{u}}(\alpha,t)}{dt} + i\left[\overline{\alpha}\,\widehat{\widetilde{u}}(\alpha,t) + \frac{\gamma+1}{2} \cdot \int_{-\infty}^{\infty} \overline{k}(k)\,\widehat{\widetilde{u}}(k,t)\,\widehat{\widetilde{u}}(\alpha-k,t)dk\right] = 0 \tag{2.19}$$

where $\overline{k}(k)$ is the wavenumber of the finite difference scheme.

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Equation (2.15) describes the time evolution of a nonlinear acoustic pulse in wavenumber space. Equation (2.19) describes the time evolution of the same pulse in wavenumber space as computed by the finite difference scheme. By comparing the solutions of these two equations, insight into the behavior of the nonlinear finite difference equation can be gained. These nonlinear integral equations can be solved numerically by first dividing the wavenumber axis into a fine mesh so that $\widetilde{u}(\alpha_j, t)$ represents the value of $\widetilde{u}$ at the jth mesh point. The integrals are then replaced by finite sums. In this way, these equations are converted into a system of ordinary differential equations which can be integrated numerically by the Runge-Kutta method. Figure 8 shows the solutions of equations (2.15) and (2.19) at $t = 6$ for an acoustic pulse the initial profile of which is given by (2.14) with $h = 0.5$ and $b = 5.0$ or (1.4). The time evolution of the pulse in physical space is provided in figure 2. At $t = 6$ the wave spectrum is smooth and nearly Gaussian as shown in figure 8a. There is very little difference between the exact and the finite difference solutions. Figure 8b shows the wavenumber spectrum at $t = 11.67$, the shock formation time for the pulse. The exact wavenumber spectrum and that computed by the finite difference scheme are no longer identical. A small spurious peak appears in the spectrum of the finite difference scheme at $\alpha\,\Delta x = 1.8$. This peak grows in height as time increases. At $t = 16$ the spectrum is given in figure 8c. The wave energy contained in the spurious peak in the wavenumber spectrum produces the unwanted spatial oscillations in the physical space, see figure 2, as noted before. The dominant wavelength of the spatial oscillations is about $3.5\Delta x$ which corresponds to a wavenumber of $\alpha\,\Delta x \simeq 1.8$.

It is clear from figure 8 that the accumulation of wave energy around the spurious peak is the cause of the appearance of the extraneous spatial oscillations in the waveform in physical space. Extensive numerical experiments have been carried out to determine the factors which would significantly influence the wavenumber α_{peak} at which the spurious peak occurs. The factors investigated include:

(1) The initial waveform of the acoustic pulse.
(2) Whether the equation is in a conservation or non-conservation form before it is discretized.
(3) The size of the finite difference stencil N.

Several families of initial waveforms have been investigated. They are (a) the Gaussian profile

$$u(x,0) = h\exp\left[-\ln 2\left(\frac{x}{b}\right)^2\right] \tag{2.20}$$

(b) The Lorentz line shape

$$u(x,0) = \frac{h}{1+(x/b)^2} \tag{2.21}$$

(c) The half-period cosine function

$$u(x,0) = \begin{cases} h\cos(\frac{\pi x}{3b}), & -\frac{3b}{2} \le x \le \frac{3b}{2} \\ 0, & \text{otherwise} \end{cases} \tag{2.22}$$

Each of these waveforms are characterized by two parameters, namely, h (the maximum amplitude) and b (the half-width). For each waveform, equation (2.19) which is the Fourier transform of the discretized form of equation (2.9) (in dimensionless form) and its counterpart

$$\frac{d\widehat{\widetilde{u}}}{dt} + i\left[\overline{\alpha}\,\widehat{\widetilde{u}} + \frac{\gamma+1}{4}\,\overline{\alpha} \cdot \int_{-\infty}^{\infty} \widehat{\widetilde{u}}(\alpha-k,t)\,\widehat{\widetilde{u}}(k,t)dk\right] = 0 \tag{2.23}$$

which is the Fourier transform of the discretized form of (2.9) rewritten in a conservation form (see equation (1.5)) are solved. The location, α_{peak}, of the spurious peak in the wavenumber space using stencil size $N = 3$, 5 and 7 are determined. The results are shown in figures 9a to 9c.

An examination of figure 9 indicates, first of all, that whether the finite difference scheme is applied to the governing partial differential equation casted in a conservation form or not has only minor effect on the location of the spurious spectrum peak. The profile shape, on the other hand, has a somewhat stronger effect. However, the stencil size or N appears to have the largest influence. The larger the stencil size, the higher is the value of $\alpha_{\text{peak}}\,\Delta x$. By concentrating on a fixed stencil size, it is easy to see that the spurious peak invariably lie near the peak region of the $\overline{\alpha}\,\Delta x$ versus $\alpha\,\Delta x$ curve in the wavenumber range as marked by the line AB shown in figure 7. The lower limit, point A, is very nearly equal to $\alpha_c\,\Delta x$ so that the wavenumber of the spurious spectrum peak is definitely in the short wave range. That the wavenumber of the spurious peak lie close to the wavenumber at which $\overline{\alpha}$ is the largest, is not surprising. The nonlinear wave steepening term of equation (2.9) is $u(\partial u/\partial x)$ or $1/2(\partial u^2/\partial x)$. This term causes the waves to pile up on the front side of the acoustic pulse creating a steep gradient. Since a steep gradient implies high wavenumbers one can regard the effect of wave steepening as causing a continuous cascade of wave energy

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into the high wavenumber range. In the wave spectrum equation (2.23) the nonlinear wave steepening term of the discretized equation is given by the last term on the left involving the convolution integral. This term is weighted by the factor $\overline{\alpha}(\alpha)$. Thus unlike the original partial differential equation, there is a built-in preference to transfer energy to the wavenumber range for which $\overline{\alpha}$ is the largest. In other words, the nonlinear finite difference equation will not transfer wave energy into finer and finer scales or higher and higher wavenumbers as in the case of the original nonlinear partial differential equation. Instead, nonlinear effect tends to cascade energy to the wavenumber range around the peak region of the $\overline{\alpha}\,\Delta x$ versus $\alpha\,\Delta x$ curve. In this peak region, the waves supported by the finite difference scheme are dispersive ($d\,\overline{\alpha}\,/d\,\alpha$ is not constant). It is well-known in wave propagation theory (see Whitham[23]) that dispersion effect tends to counteract nonlinear steepening effect so that once the wave energy is transferred to this range they will stay in this range. In the physical space, the dispersive waves propagate at a slower speed than the main pulse (smaller group velocity). They are left behind the pulse so that for these waves nonlinear effects no longer apply and their wavenumbers remain unchanged with time.

Finally, the wavenumber at which $\overline{\alpha}\,\Delta x$ (the wavenumber of the finite difference stencil) attains its peak value, is larger for larger size stencil as illustrated in figure 7. The above reasoning, therefore, suggests that the spurious spatial oscillations which appear in the waveforms of finite difference computations, would have shorter wavelength if a larger stencil is used. This is consistent with all the numerical experiments performed in this work.

3. Artificial Selective Damping

We have seen that the most objectionable consequence of using a high-order finite difference scheme to calculate nonlinear acoustic wave pulse solution is the inevitable appearance of spurious spatial oscillations behind and sometimes in front of a shock. However, we have also found that these oscillations are concentrated in a relatively narrow range of wavenumber around the value $\alpha = \alpha_{\text{peak}}$. This observation suggests that if we can eliminate or damp out the waves in this narrow range of wavenumbers by adding artificial damping terms to the governing equations but leaving the other wavenumbers undamped, we would effectively remove the extraneous spatial oscillations. In this way, an acceptable wave profile could be found in the use of high-order finite difference schemes.

The idea of using artificial damping terms to smooth out the profile of a shock is not new. It was first suggested by Von Neumann and Richmyer[24] and used very effectively by Jameson *et.al*[25] and others. The advantage we have here is that we know for a given computation stencil size the range of wavenumber over which damping needs to be concentrated; more or less regardless of the initial pulse shape and whether the equation is put in conservation form to begin with. The significant range where artificial damping is to be applied for the 7-point, 11-point and 15-point stencils are provided in figure 7. Of course, to protect the quality of the computed solution, there should be minimal damping in the long wave range (wavenumbers below α_c or the point A in figure 7). The above consideration suggests that constant damping over all wavenumber should not be used. What is needed is to incorporate into the governing equation selective damping terms which have extremely small damping over the long wave range but significant damping in the short wave range. Tam, Webb and Dong[8] have recently considered such artificial selective damping terms. They showed how the coefficients of the damping terms can be chosen to control the range of wavenumber where significant damping may be realized. We will now first examine some of the effects of selective damping on the propagation of acoustic pulse. In the next section, we will discuss how such damping terms can be used effectively for nonlinear wave computation by high-order finite difference schemes.

Suppose artificial damping terms, $\overline{D}$, are added to the right side of the convective wave equation (1.1) written in dimensional form;

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \overline{D} \tag{3.1}$$

where c is the wave speed. Upon discretizing (3.1) and assuming that the damping terms are proportional to the values of u at the neighboring mesh points within the stencil, the following governing equation for the ℓth mesh point is obtained,

$$\frac{du_\ell}{dt} + \frac{c}{\Delta x}\sum_{j=-3}^{3} a_j u_{\ell+j} = -\frac{\nu_a}{(\Delta x)^2}\sum_{j=-3}^{3} c_j u_{\ell+j} \tag{3.2}$$

In (3.2) $\nu_a/(\Delta x)^2$ is the damping coefficient. ν_a is the artificial kinematic viscosity. Δx is the mesh spacing. On generalizing (3.2) to a difference equation with continuous variable in x and then taking the Fourier transform of the equation, it is easy to find

$$\frac{d\,\widetilde{u}}{dt} + c\,\overline{\alpha}\,\widetilde{u} = -\frac{\nu_a}{(\Delta x)^2}D(\alpha\,\Delta x)\,\widetilde{u} \tag{3.3}$$

where

$$D(\alpha\,\Delta x) = \sum_{j=-3}^{3} c_j e^{ij\,\alpha\,\Delta x} \tag{3.4}$$

From (3.3) it is clear that $D(\alpha\,\Delta x)$ represents the (relative) damping rate for the wavenumber α. Tam, Webb and Dong[8] showed how the coefficients c_j could be chosen so that there is relatively little damping for small $\alpha\,\Delta x$; i.e., long waves but that there is significant damping for high wavenumbers (short waves). Three conditions on the coefficients c_j are self-evident. First is that D should be a positive even function of $\alpha\,\Delta x$. The even function property is assured by setting $c_{-j} = c_j$. Thus

$$D = c_0 + 2\sum_{j=1}^{3} c_j \cos(j\,\alpha\,\Delta x) \tag{3.5}$$

Second, there should be no damping for very long waves. That is, $D \to 0$ as $\alpha \to 0$. This requires

$$\sum_{j=-3}^{3} c_j = 0 \tag{3.6}$$

Thirdly, D may be normalized so that $D(\pi) = 1.0$.

The right side of (3.5) is a truncated Fourier series. To ensure that $D(\alpha\,\Delta x)$ is small when $\alpha\,\Delta x$ is not too large but becomes large when $\alpha\,\Delta x \to \pi$, Tam, Webb and Dong suggested to use the Gaussian function centered at π with half-width σ as a template for choosing the Fourier coefficients c_j. This is done by determining c_j such that the integral

$$\int_0^{\beta} \left\{ D(\alpha\,\Delta x) - \exp\left[-\ln\,2\left(\frac{\alpha\,\Delta x - \pi}{\sigma}\right)^2\right]\right\}^2 d(\alpha\,\Delta x)$$

is minimum. The upper limit of the integral, β, is a parameter which can be adjusted to yield the most desirable property for D. For example, with $\sigma = 0.3\pi$, $\beta = 0.65\pi$, it is easy to find

$$c_0 = 0.327698660846$$
$$c_1 = -0.235718815308$$
$$c_2 = 0.086150669577$$
$$c_3 = -0.014281184692$$

On the other hand, with $\sigma = 0.2\pi$, $\beta = 0.33327\pi$ the coefficients are found to be

$$c_0 = 0.303058855153$$
$$c_1 = -0.231695528614$$
$$c_2 = 0.098470572424$$
$$c_3 = -0.018304471386$$

The damping curves, D as a function of $\alpha\,\Delta x$, for the above examples are shown in figure 10. D is very small over the small wavenumber range. It is significantly larger for $\alpha\,\Delta x > 1.4$. These are the desired damping characteristics.

What are the effects of artificial selective damping on the computed acoustic pulse solution? Obviously, the damping terms would reduce the amplitude of the pulse over time. However, there are some not so obvious side effects. Figure 11 shows the computed solution of equation (3.2) and initial condition (1.2) after the pulse has propagated over a distance of 400 mesh points. In the computation, dimensionless variables with Δx as the length scale, c as the velocity scale and $\Delta x/c$ as the time scale were used. The artificial Reynolds number $c\Delta x/\nu_a$ was taken to be 0.2 and the $\sigma = 0.3\pi$ damping curve was included on the right side of the equation. The exact solution without artificial selective damping is shown as the dotted curve in figure 11. The solid curve represents the computed waveform with artificial selective damping included. By comparing the two waveforms, it is apparent that there is overall reduction in the wave amplitude. However, at points farther away from the center of the pulse, the wave amplitude has actually increased as if the pulse has dispersed outwards on both sides. The cause of this apparent dispersion effect is subtle but can be easily understood by viewing the damping process in the wavenumber space. Since the Fourier transform of a Gaussian function is also a Gaussian function, the initial pulse is also a concentrated pulse around $\alpha\,\Delta x = 0$ in the wavenumber space. Now, the artificial selective damping terms gradually reduce the pulse amplitude but not evenly. By design, there is no reduction at zero wavenumber. The reduction increases with increase in wavenumber. Because of this the half-width of the pulse in wavenumber space is reduced over time. The maximum height which occurs at zero wavenumber, nevertheless, remains the same. The waveform in the physical space is the inverse transform of the waveform in the wavenumber space. With a narrower pulse, as time increases, the physical waveform spreads out according to the inverse spreading theorem of Fourier transform (a narrow pulse has a Fourier transform of larger half-width). Thus artificial selective damping has the somewhat unintended effect of causing numerical dispersion.

4. Variable Artificial Damping

The purpose of adding artificial damping terms to the difference equation is to eliminate the spurious spatial oscillations generated by nonlinear steepening

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effect. Clearly then, artificial damping is needed only in the limited regions of space where large gradients occur. The use of constant artificial viscosity which introduces the same amount of damping everywhere is, therefore, not a good strategy. For high quality numerical results, a variable artificial viscosity appears to be desirable.

To establish a variable damping algorithm, a way to measure the shock strength or velocity gradient is needed. A shock is expected to be smeared out over a few mesh points, so the relevant measure of length is Δx (the mesh size). As a simple measure of velocity we propose to use $u_{\text{stencil}} = |u_{\max} - u_{\min}|$; the difference between the maximum velocity, $u_{\max}$, in the stencil and the minimum velocity, $u_{\min}$, in the stencil. (Note: If a very large finite difference stencil is used in the computation one might consider restricting the determination of $u_{\max}$ and $u_{\min}$ to a smaller stencil in forming u_{stencil}.) We will define the stencil Reynolds number as,

$$R_{\text{stencil}} = \frac{u_{\text{stencil}}\Delta x}{\nu_a}$$

The Euler equations in two dimensions may be written in the form,

$$\mathbf{A}\frac{\partial \mathbf{u}}{\partial t} + \mathbf{B}\frac{\partial \mathbf{u}}{\partial x} + \mathbf{C}\frac{\partial \mathbf{u}}{\partial y} = 0 \tag{4.1}$$

where

$$\mathbf{u} = \begin{bmatrix} \rho \\ u \\ v \\ p \end{bmatrix}, \ \mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \rho & 0 & 0 \\ 0 & 0 & \rho & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} u & \rho & 0 & 0 \\ 0 & \rho u & 0 & 1 \\ 0 & 0 & \rho u & 0 \\ 0 & \gamma p & 0 & u \end{bmatrix}, \ \mathbf{C} = \begin{bmatrix} v & 0 & \rho & 0 \\ 0 & \rho v & 0 & 0 \\ 0 & 0 & \rho v & 1 \\ 0 & 0 & \gamma p & v \end{bmatrix}$$

Upon discretizing (4.1) over a uniform mesh of spacings $\Delta x = \Delta y$ and incorporating artificial damping terms of the form given in equation (3.2), the Euler equations become,

$$\begin{aligned} &\mathbf{A}_{\ell m}\frac{d\mathbf{u}_{\ell m}}{dt} + \mathbf{B}_{\ell m}\sum_{j=-3}^{3}\frac{1}{\Delta x}a_j\mathbf{u}_{\ell+j,m} \\ &\quad + \mathbf{C}_{\ell m}\sum_{j=-3}^{3}\frac{1}{\Delta y}a_j\mathbf{u}_{\ell,m+j} \\ &= -\frac{u_{\text{stencil}}}{\Delta x R_{\text{stencil}}}\mathbf{A}_{\ell,m}D(\mathbf{u}_{\ell m}) \end{aligned} \tag{4.2}$$

where

$$D(\mathbf{u}_{\ell m}) = \sum_{j=-3}^{3} c_j(\mathbf{u}_{\ell+j,m} + \mathbf{u}_{\ell,m+j}) \tag{4.3}$$

If the problem is one-dimensional involving disturbances propagating into a region originally at rest then the Euler equations reduce to the nonlinear simple wave equation[23] (see equation (2.9)). The discretized form of this equation including artificial damping terms is

$$\begin{aligned} &\frac{du_\ell}{dt} + \left(a_0 + \frac{\gamma+1}{2}u_\ell\right)\frac{1}{\Delta x}\sum_{j=-3}^{3}a_j u_{\ell+j} \\ &= -\frac{\gamma+1}{2}\frac{u_{\text{stencil}}}{\Delta x R_{\text{stencil}}}\sum_{j=-3}^{3}c_j u_{\ell+j} \end{aligned} \tag{4.4}$$

In the present formulation, the stencil Reynolds number has the meaning of the ratio of the magnitude of the nonlinear steepening terms to that of the artificial damping terms. If the two effects are in balance, namely, the rate of production of short waves by the nonlinear steepening terms is equal to the rate at which they are removed by artificial damping, then one would expect this ratio to be a universal constant; provided a good quality damping curve is used. Here a universal constant means that it is independent of the acoustic problems to be solved. Its value obviously depends on the finite difference scheme and the damping curve. Extensive numerical experiments using the $\sigma = 0.3\pi$ damping curve have been carried out on equation (4.4) and the one-dimensional version of (4.2). It is found that by taking $R_{\text{stencil}} = 0.1$ most pre-shock and post-shock spatial oscillations are removed from the numerical solutions for all the initial pulse profiles we have considered. Figure (12) shows the numerical solution of (4.4) with an initial waveform given by (2.14). Dimensionless variables with Δx, a_0 and $\Delta x/a_0$ as length, velocity and time scales are used in the computation. The pulse is narrow with $h = 0.5$ and $b = 3.0$. The computed solution appears to be free of spurious spatial oscillations and agrees well with the exact solution. The shock has been smoothed out over a distance of 4 to 5 mesh spacings. Figure 13 shows a similar comparison with exact solution. In this case, the height and width of the initial pulse shape are $h = 1.0$ and $b = 12.0$, respectively. Again, the variable artificial damping with $R_{\text{stencil}} = 0.1$ has removed essentially most of the spurious spatial oscillations. The agreement with the exact solution is reasonably good.

In a recent work, Yang and Przekwas[20] tested and compared the shock capturing capability of sixteen

numerical schemes by applying them to the Burger's equation. Two initial value problems were solved using each of the schemes. The first problem involves an initial parabolic profile which steepens into two shocks. The second problem involves expansion fans and collision of shocks. The Burger's equation does not give rise to long distance wave propagation so that the dispersion property of these schemes was not rigorously tested. To evaluate the effectiveness of the present variable artificial damping methodology relative to these other schemes, the Burger's equation with artificial damping terms added; i.e.,

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = -\frac{u_{\text{stencil}}}{\Delta x R_{\text{stencil}}} D \tag{4.5}$$

is solved computationally by the DRP scheme. The stencil Reynolds number, R_{stencil}, is again set equal to 0.1. Figures 14a and 14b show the computed and the exact solutions at $t = 2$ and $t = 12$ for the first problem. Figures 15a and 15b show the computed and the exact solutions at $t = 0.75$ and $= 2.0$ for the second problem. These results are to be compared with those given by Yang and Przekwas using other schemes. Obviously the present results are not as good as some of the schemes designed specifically for capturing shocks. On the other hand, the present results are quite good compared to the results of many of the remaining schemes. Judging from all the above numerical results, we believe that the adding of variable artificial damping terms to a high-order finite difference scheme, such as the DRP scheme, allows the scheme to be used for nonlinear acoustic wave propagation computation. Such a high-order scheme has little numerical dispersion (for long waves), possesses the correct wave speed and other wave propagation characteristics and generates little spurious spatial oscillations even when the acoustic pulse steepens into a shock wave.

5. Summary and Discussion

In this work, the advantages and disadvantages of using high-order finite difference schemes to compute nonlinear acoustic pulse solutions are investigated. The advantages are two-fold. First, when properly formulated, these schemes are less dispersive, less dissipative, more isotropic and more capable of reproducing the correct speed of propagation than lower order schemes. Second, these schemes can better approximate the solution of the original partial differential equation over a much larger range of wavenumbers. The main disadvantages are the tendency of the schemes to generate spurious spatial oscillations around a shock and the smearing out of the shock over a larger number of mesh points. The origin of the spurious spatial oscillations has been investigated. It is linked to the inability of the finite difference scheme to faithfully approximate the short wave components. The spurious oscillations are confined to a narrow range of wavenumbers. To remove these spurious oscillations, it is proposed that artificial selective damping terms specifically targeting the short wave range be added to the governing equations. Since large damping is required only in regions where large velocity gradients (and hence nonlinear steepening effects) exist, it is suggested that a variable damping coefficient be used. Consideration of attaining a balance between the nonlinear steepening effect and the artificial damping effect leads to the concept of a universal stencil Reynolds number. That is, the value of the stencil Reynolds number depends only on the finite difference scheme and the damping curve used. It is independent of the acoustic problems to be solved. Extensive numerical experiments based on the 7-point DRP scheme and a damping curve with a half-width of 0.3π indicate that the use of $R_{\text{stencil}} = 0.1$ generally results in a reasonably good nonlinear acoustic pulse profile which is free from most spurious spatial oscillations.

For simplicity, the present work has, so far, been concentrated on one-dimensional acoustic wave problems. Efforts are now underway to extend the investigation to two-dimensional Euler equations. Both initial and time periodic problems will be considered. In addition, a study of the use of artificial damping terms in conjunction with the DRP finite difference scheme of very large stencil ($N = 5$ and 7) is planned. In these cases it is anticipated that a damping curve of narrower half-width may be sufficient. The significance of these very high-order finite difference schemes is that they provide a wider long wave range for numerical computation. In other words, they can resolve acoustic wave component of much shorter wavelengths.

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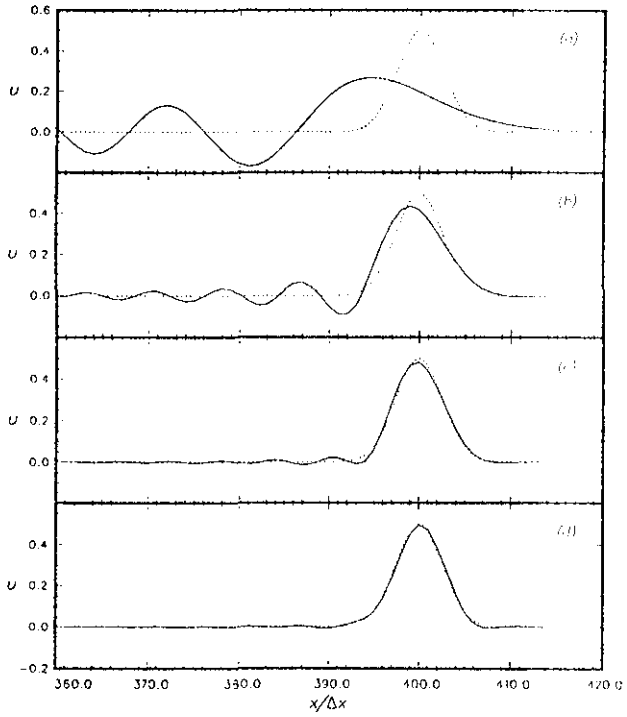

Figure 1. Comparison between computed and exact solutions of the simple one dimensional wave equation. ——— numerical solution; ····· exact solution. (a) Second order central difference scheme. (b) Fourth order central difference scheme (c) Sixth order central difference scheme. (d) DRP scheme (7-point stencil).

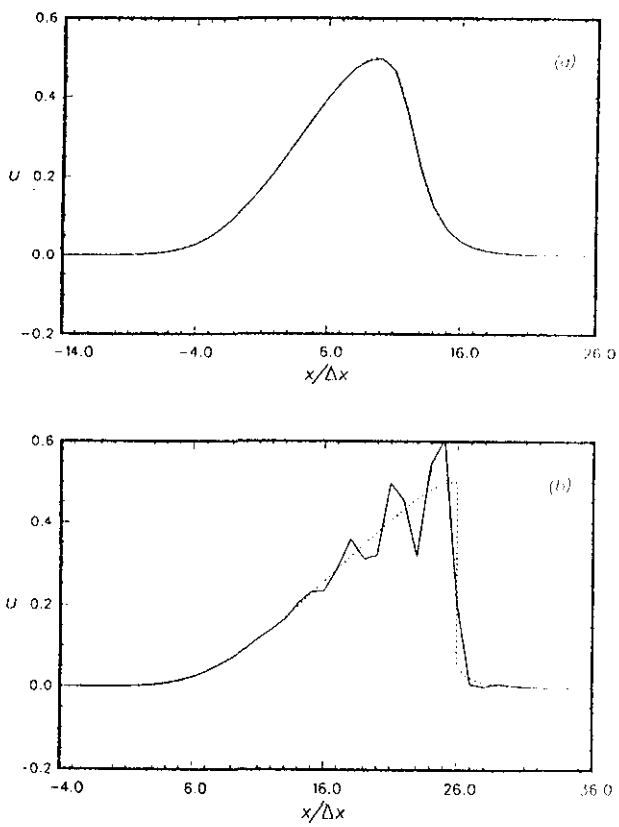

Figure 2. Comparison between computed and exact solutions of the nonlinear simple wave equation. ····· exact solution. ——— DRP scheme (7-point stencil) in nonconservation form (a) t = 6.0, (b) t = 16.0.

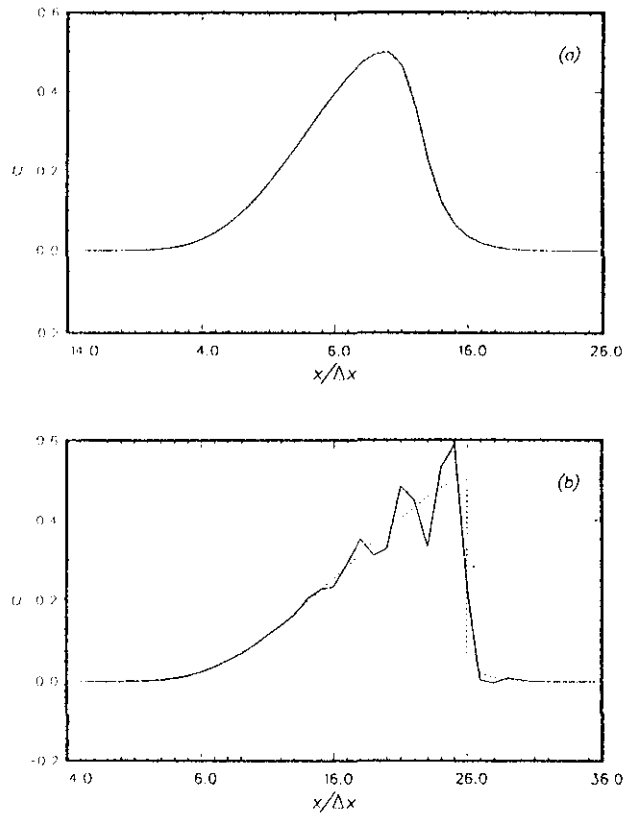

Figure 3. Comparison between computed and exact solutions of the nonlinear simple wave equation. ····· exact solution. ——— DRP scheme (7-point stencil) in conservation form (a) t = 6.0, (b) t = 16.0.

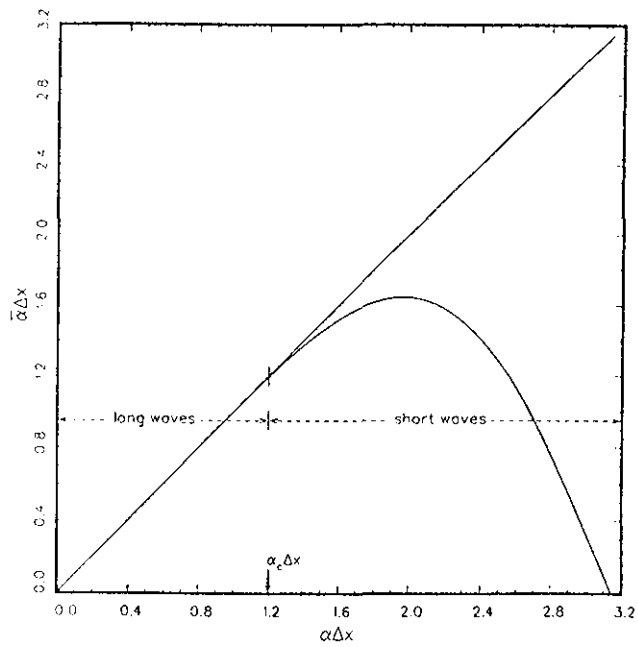

Figure 4. $\bar{\alpha}\Delta x$ versus $\alpha\Delta x$ curve for the optimized (−1.1 to 1.1) 7-point stencil (N=3) Dispersion–Relation–Preserving scheme showing the long and short wave range.

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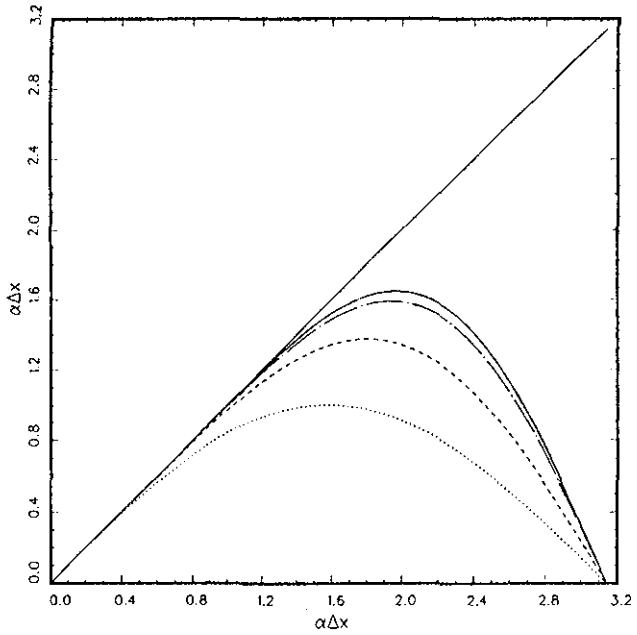

Figure 5. $\bar{\alpha}\Delta x$ versus $\alpha\Delta x$ relation for the standard central difference second order, ·····; fourth order, - - - - -; sixth order, —·—; and the DRP scheme, ——.

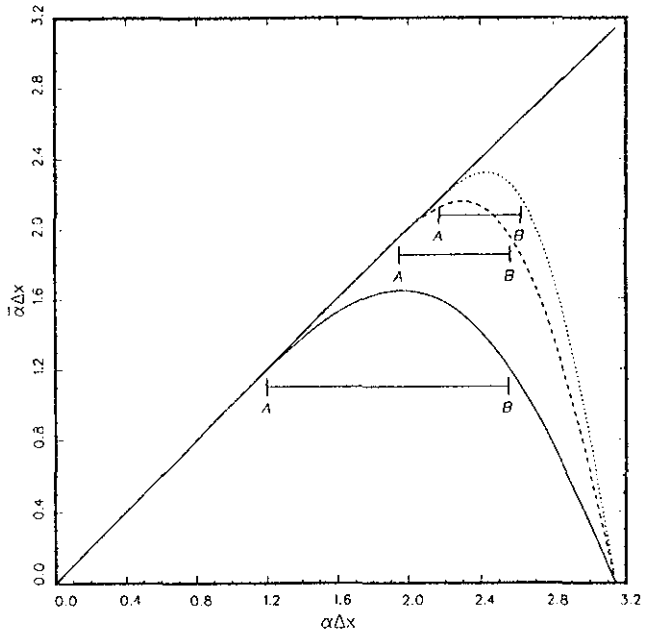

Figure 7. $\bar{\alpha}\Delta x$ versus $\alpha\Delta x$ relation for the DRP schemes.
—— N=3; - - - - - N=5; ····· N=7.

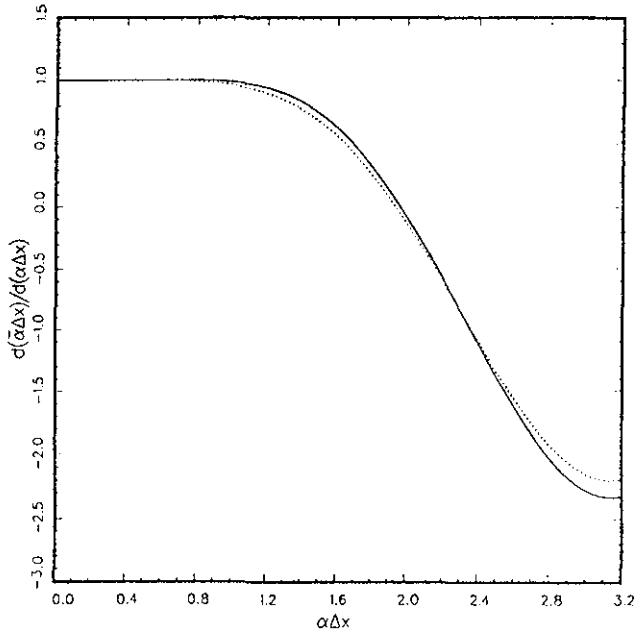

Figure 6. Comparison between the $d\bar{\alpha}/d\alpha$ versus $\alpha\Delta x$ curves for the standard sixth order central difference scheme, ····· and the DRP scheme, ——.

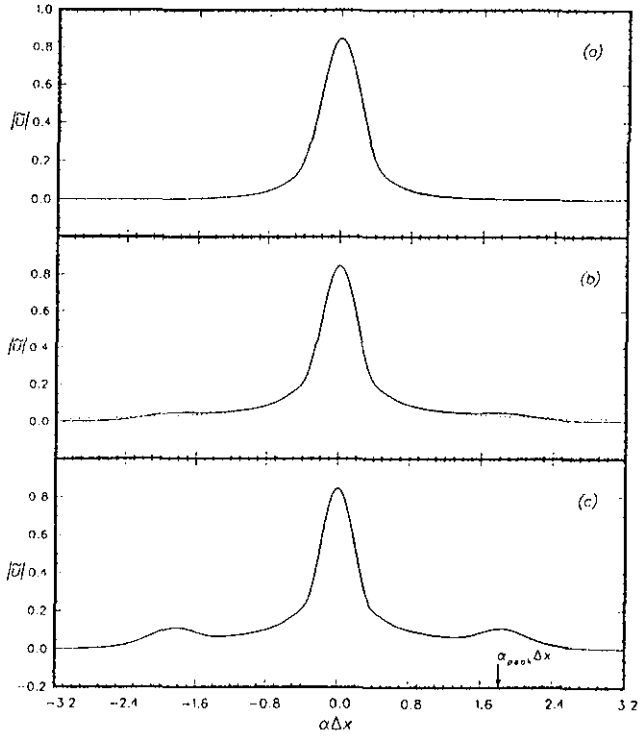

Figure 8. Time evolution of the wavenumber spectrum of an acoustic pulse. —— DRP scheme, ····· exact. (a) t = 6.0, (b) t = 11.67, (c) t = 16.0.

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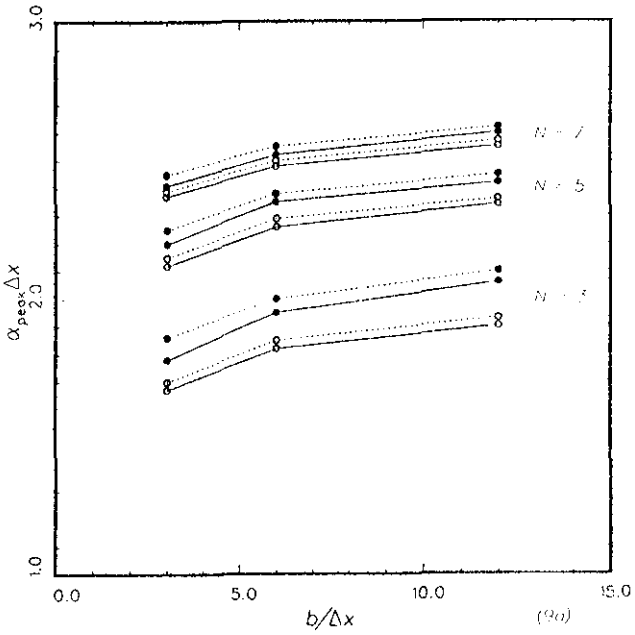
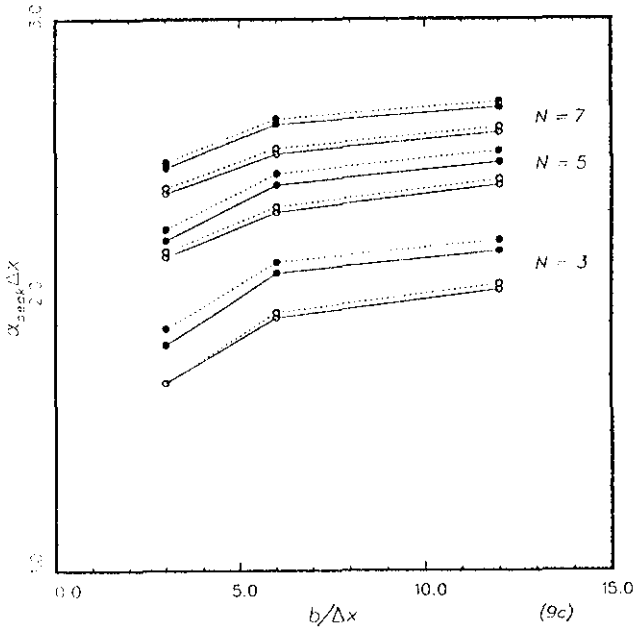
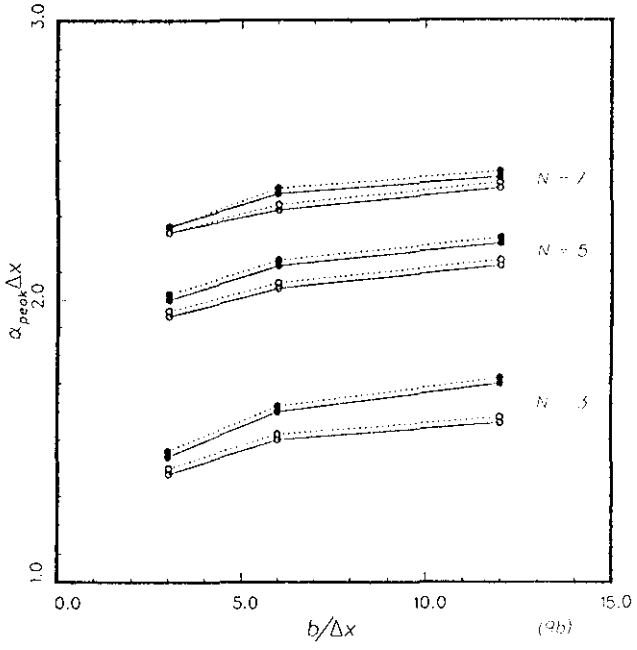


Figure 9. Variation of $\alpha_{peak}\Delta x$ on initial pulse profile parameters, stencil size and the use of the equation in conservation or nonconservation form. (a) Gaussian profile, (b) Lorentz line shape, (c) Single half period cosine function. —— *equation in conservation form,* ····· *equation in nonconservation form.*
∘ *h = 0.5,* • *h = 1.0.*





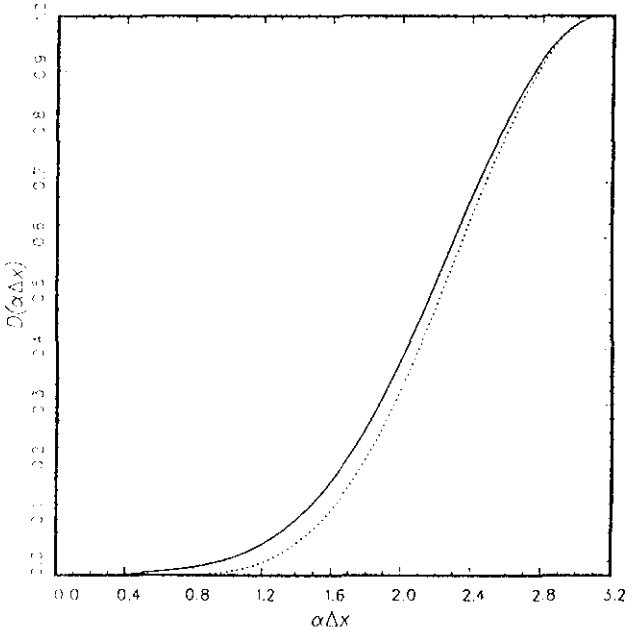


Figure 10. The damping curves. —— $\sigma = 0.3\pi$; ····· $\sigma = 0.2\pi$.

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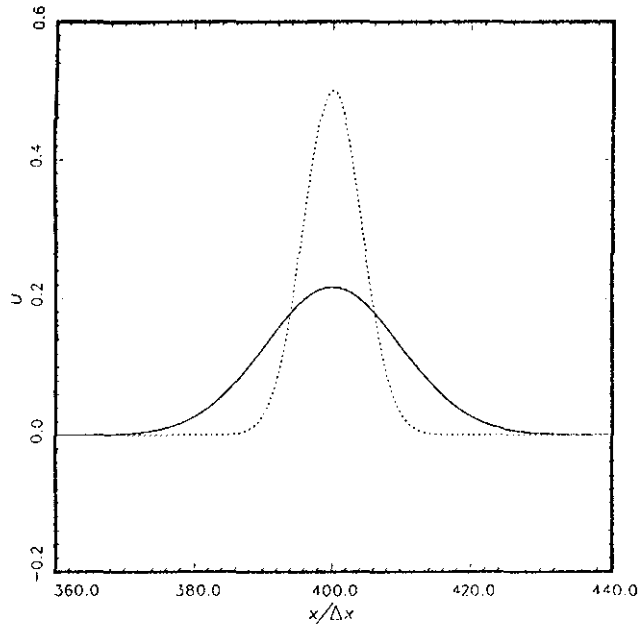

Figure 11. Effects of artificial damping on an acoustic pulse. ····· waveform without damping; ——— waveform with damping.

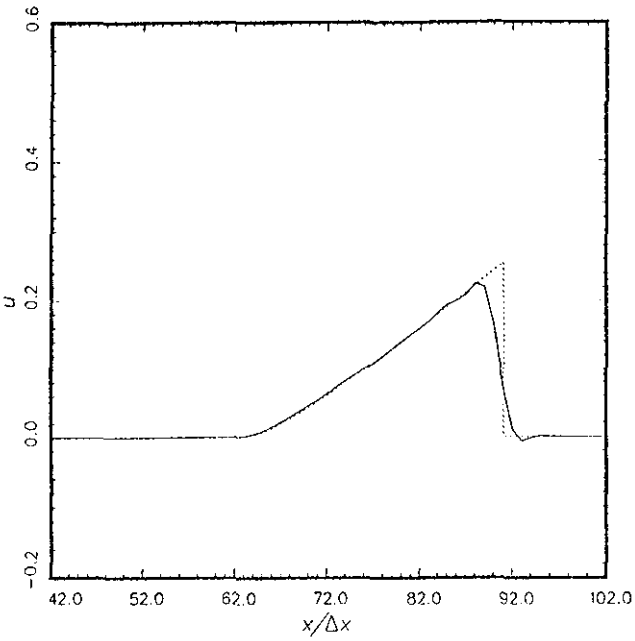

Figure 12. Computation of nonlinear acoustic pulse with variable artificial damping. $R_{stencil} = 0.1$, $t = 72$. Gaussian initial waveform with $h = 0.5$ and $b = 3\Delta x$. ——— numerical solution; ····· exact solution.

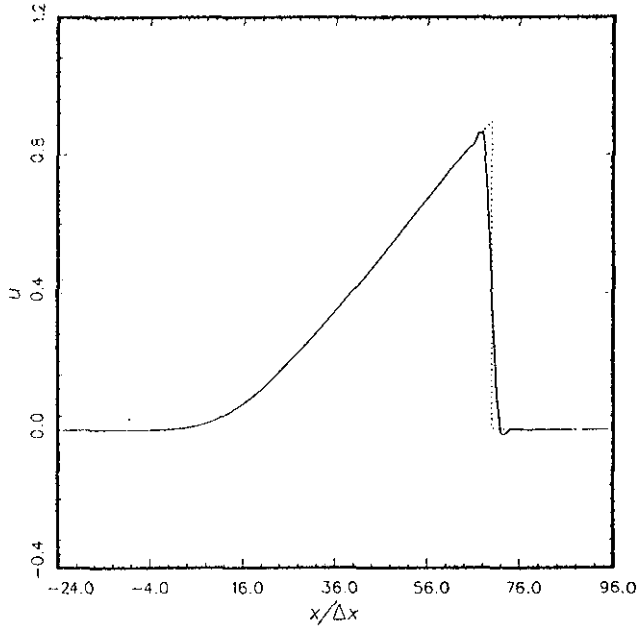

Figure 13. Computation of nonlinear acoustic pulse with variable artificial damping. $R_{stencil} = 0.1$, $t = 36$. Gaussian initial waveform with $h = 1.0$ and $b = 12\Delta x$. ——— numerical solution; ····· exact solution.

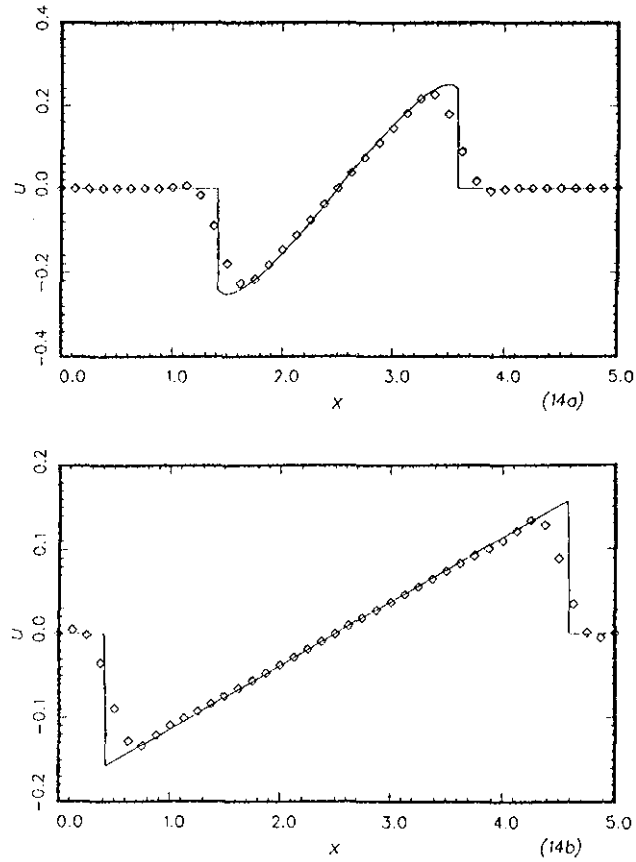

Figure 14. Comparison between numerical solution of the Burgers' equation with variable artificial damping and the exact solution. $R_{stencil} = 0.1$. Parabolic initial profile. (a) $t = 2.0$, (b) $t = 12.0$.

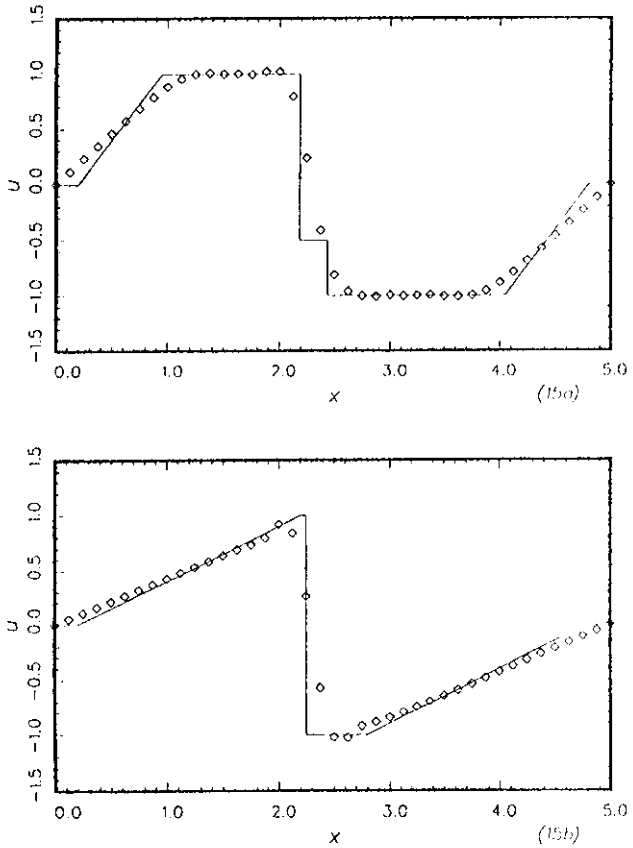

Figure 15. Comparison between numerical solution of the Burgers' equation with variable artificial damping and the exact solution. $R_{stencil} = 0.1$. Collision shocks problem. (a) $t = 0.75$, (b) $t = 2.0$.

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